E 4.05 – Answers 2021

Question 1)

EE4.05 2021 1/9

a) $N = 20$

$\frac{1}{M} = 120\ s$, $\alpha = 300$ calls/h $\Big\}$ $\rho = 10$ from table: $B_C = 0.002$

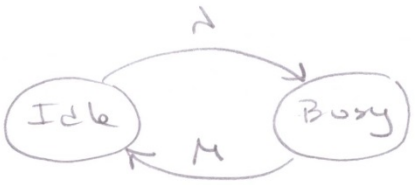
i) $\rho_C = 10(1 - 0.002) = 9.98$ erlangs

ii) $Var = \rho_C - \rho_0 B_C [N - \rho_C] = 9.59$

iii) $\eta = \frac{\rho_C}{N} = \frac{9.98}{20} = 0.499$

b)

i) source model



$P(\text{source busy}) = \frac{\lambda}{\lambda + \mu} = p$

ii) $M \leq N \quad \pi_i = \binom{M}{i} p^i (1-p)^{M-i}$

iii) $M > N \quad \pi_i = \frac{\binom{M}{i} p^i (1-p)^{M-i}}{\sum_{j=0}^{N} \binom{M}{j} p^j (1-p)^{M-j}}, \quad i = 0, \ldots, N$

(truncated distribution)

iv) $M >> N$

$\lambda_i = \left(1 - \frac{i}{M}\right)\lambda_0 \sim \lambda_0$ for all i

This model became the Erlang model.

Question 2)

(2) 2/9

a) $N = 14$

offered traffic = 11.2 erlangs

$\lambda = 288$ calls/h — 0.08 calls/s

$$\rho = \left(\frac{0.08 \times 140}{14}\right) = 0.8$$

$E_{14}(11.2) \approx 0.1$

i) $$P(\text{Delay}) = \frac{E_K(K\rho)}{1-\rho[1-E_K(K\rho)]} = \frac{0.1}{1-0.8[1-0.1]}$$

$$= 0.357$$

ii) $$E[W|W>0] = \frac{h}{K(1-\rho)} = \frac{140}{14[1-0.8]} = 50\ s$$

b)

i) $$E[Q_t|W>0] = \frac{\rho}{1-\rho} - \left(\frac{B\rho^B}{1-\rho^B}\right) \qquad \rho = 0.8,\ B = 10$$

$$= 2.79$$

ii) $$E[W|\text{Delay}] = \frac{E[Q_t|W>0]}{\lambda_E}$$

$$\lambda_E = \lambda[1-P(\text{loss})]$$

3/9

$$P(\text{loss}) = \left[\frac{(1-\rho)\rho^B \, E_K(k\rho)}{(1-\rho) + \rho(1-\rho^B) E_K(k\rho)} \right]$$

$B = 10 \quad \rho = 0.8 \quad E_K(k\rho) = E_{14}(11.2) \approx 0.1$

$P(\text{loss}) = 0.0079$

$\lambda_E = 0.08 \cdot [1 - 0.0079] = 0.0793$

$E[W | \text{Delay}] = \frac{2.79}{0.0793} = 33.18 \text{ s}$

iii) compare and discuss results in parts 2a ii) and 2b ii)

Question 3)

(3)

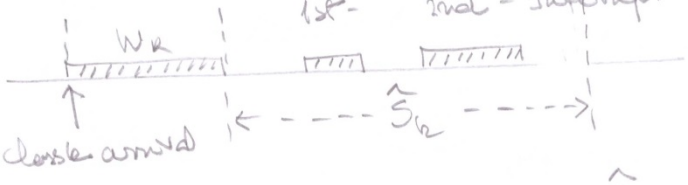
i)

4/9

- Definition and comparison between preemptive resume and preemptive restart priority
- A preemptive priority system could be analysed using mean value analysis (MVA).
- solve for preemptive resume priority and then bound preemptive restart priority

ii)

start preemptive resume



define Transit time $T_k = W_k + \hat{S}_k$

V_k = work into the system during $\hat{S}_k$ by higher priority arrivals.

$$E[V_K] = \sum_{i=1}^{k-1} (\lambda_i E[\hat{S}_k]) E[S_i] = \left(\sum_{i=1}^{k-1} \rho_i \right) E[\hat{S}_k]$$

$$= \sigma_{k-1} E[\hat{S}_k]$$

$$E[\hat{S}_k] = E[S_k] + E[V_k] \Rightarrow E[\hat{S}_k] = \frac{E[S_k]}{1 - \sigma_{k-1}}$$

$$E[T_k] = E[W_k] + E[\hat{S}_k]$$

This is a bound to $E[T_k]$ preemptive restart for preemptive restart $E[\hat{S}_k]$ will be longer

(3)

i) Y_t = state of the underlying ON-OFF switching process

N_t = Number of busy channels in overflow link

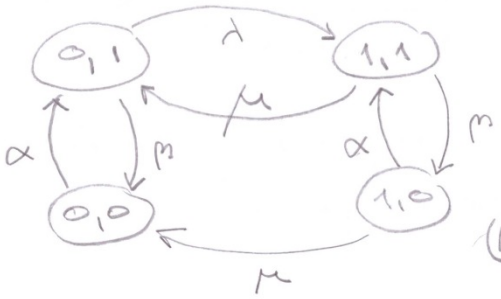
- only one overflow link

$\lambda = 180$ calls/h $= 3$ calls/m

$1/\mu = 180$ s $= 3$ m $\quad \mu = 1/3$

$\alpha = 0.2\ m^{-1}$

$\beta = 7.0\ m^{-1}$



$\pi_0 = (0,0) \quad \pi_1 = (1,0)$

$\sigma_0 = (0,1) \quad \sigma_1 = (1,1)$

ii) For the IPP model:

$$m = E(N_t) = \left(\frac{\alpha}{\alpha+\beta}\right)p = \frac{0.2}{7.2} \cdot 9 = 0.25$$

$$v = \text{Var}(N_t) = m\left[1 + \frac{(p/\alpha)\,m}{\alpha+\beta+1}\right]$$

$$= 0.5167$$

iii) global balance equations and $\underline{\pi}^T Q = \underline{0}^T$ 6/9

$$(\pi_0\ \sigma_0\ \pi_1\ \sigma_1)\begin{bmatrix} -\alpha & \alpha & \phi & \phi \\ \beta & -(\beta+\lambda) & \phi & \lambda \\ \mu & \phi & -(\alpha+\mu) & \alpha \\ \phi & \mu & \beta & -(\beta+\mu) \end{bmatrix}$$

with $\delta = \left(\frac{\lambda}{\mu+\alpha+\beta}\right)$

$$\pi_0 + \sigma_0 = \left[\frac{\alpha+\beta(1+\delta)}{\alpha(1+\beta_0)+\beta(1+\delta)}\right] = 0.847$$

$$\pi_1 + \sigma_1 = \left[\frac{\alpha\beta_0}{\alpha(1+\beta_0)+\beta(1+\delta)}\right] = 0.153$$

P(overflow channel busy) $= \pi_1 + \sigma_1 =$ Time congestion

Time congestion $= 0.153$

call congestion = P(overflow channel busy / arrivals on)

$$= P(N_t = 1 \mid y_t = 1) = \frac{\sigma_1}{\sigma_0+\sigma_1}$$

use also:

$$P(\text{overflow traffic ON}) = \frac{\alpha}{\alpha+\beta} = \sigma_0 + \sigma_1 = 0.028$$

$$P(\text{overflow traffic OFF}) = \frac{\beta}{\alpha+\beta} = \pi_0 + \pi_1 = 0.972$$

and:

$$\alpha \pi_0 = \beta \sigma_0 + \mu \pi_1$$

$$(\beta+\lambda)\sigma_0 = \alpha \pi_0 + \mu \sigma_1$$

$$(\alpha+\mu)\pi_1 = \beta \sigma_1$$

$$(\beta+\mu)\sigma_1 = \alpha \pi_1 + \lambda \sigma_0$$

$\pi_0 + \sigma_0 = 0.897$

$\pi_1 + \sigma_1 = 0.153$

$\sigma_0 + \sigma_1 = 0.028$

$\pi_1 + \pi_0 = 0.972$

$\alpha = 0.2$, $\beta = 7.0$, $\mu = \frac{1}{3}$, $\lambda = 3$

$\sigma_1 = 0.011$

$\pi_1 = 0.142$

$\sigma_0 = 0.017$

$\pi_0 = 0.830$

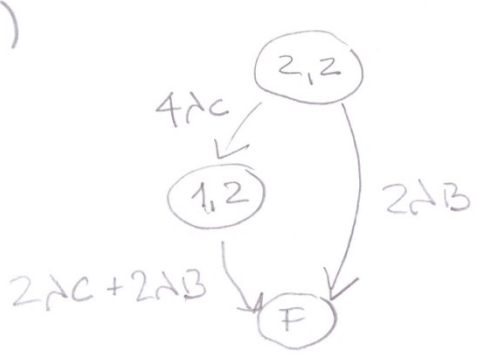
so call congestion $= \frac{\sigma_1}{\sigma_0 + \sigma_1} = 0.393$

Question 4)

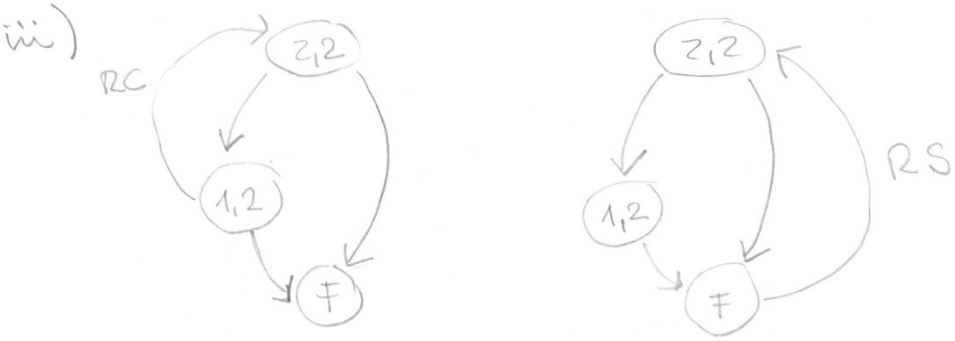
8/9

i) $S = (\text{subsystems up}, \text{buses up})$

ii)



iii)

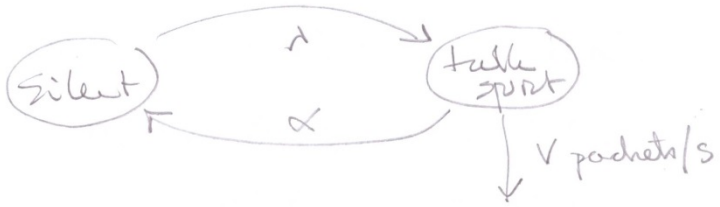


Rc = sub system replacement

RS = Complete system including buses replacement

9/9

i) C = normalised capacity of the system
N = N independent sources
$1/\alpha$ = average talk spurt
$1/\lambda$ = average silence interval.



ii) $\frac{dF(x)}{dx} D = F(x) M \rightarrow \frac{dF(x)}{dx} = F(x) M'$

$$M = \begin{bmatrix} -\lambda & \lambda \\ \alpha & -\alpha \end{bmatrix} = \begin{bmatrix} -\gamma & \gamma \\ 1 & -1 \end{bmatrix} \quad \gamma = \frac{\lambda}{\alpha}$$

$$M' = \begin{bmatrix} \frac{\gamma}{C} & \frac{\gamma}{1-C} \\ \frac{-1}{C} & \frac{-1}{1-C} \end{bmatrix} \qquad z\bar{I} - M' = 0$$

$$z = \frac{\gamma}{C} - \frac{1}{1-C}$$

condition for stable solution:

$$z < 0 \rightarrow \frac{\gamma}{C} - \frac{1}{1-C} < 0$$

$$\left(\frac{\gamma}{1+\gamma}\right)\frac{1}{C} < 1$$

$$\left(\frac{\lambda}{\lambda+\alpha}\right) < C \leftarrow \text{capacity}$$

↖ talk spurt

Discuss what would happen for N = N